

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I NITHISH KUMAR K(192521266) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: knithishkumar

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

1. Company with Seasonal Traffic Spikes

Cloud Auto-Scaling and Load Balancing

- **Auto-scaling** automatically adds or removes cloud servers based on CPU usage or user traffic. During seasonal sales, more servers are added automatically, and after the sale they are removed, reducing costs.
- **Load balancing** distributes incoming user requests evenly across all available servers, preventing any single server from becoming overloaded.

Reducing Downtime and Improving User Experience

- Auto-scaling ensures enough resources are always available during traffic spikes.
- Load balancing prevents server crashes by sharing the workload.
- Users experience faster page loading, quick checkout, and minimal downtime.

Role of Monitoring Tools

- Monitoring tools continuously track CPU, memory, network traffic, and response time.
- They trigger auto-scaling when predefined thresholds are reached.
- They send alerts if system performance degrades.

Real-World Example

During **Amazon** or **Flipkart Big Billion Days** sales, millions of customers access the website simultaneously. Auto-scaling adds new servers, while load balancing distributes customer requests, ensuring uninterrupted shopping.

Conclusion

Auto-scaling, load balancing, and monitoring together provide **high availability, better performance, reduced downtime, and cost efficiency.**

2. Secure Cloud Backup Storage

Durability and Data Redundancy

- Cloud storage keeps multiple copies of data across different servers or regions.
- If one storage device fails, another copy is automatically used.
- This ensures high durability and availability.

Encryption and Access Control

- **Encryption:** Data is encrypted during transmission (TLS) and while stored (AES-256).
- **Access Control:** Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) ensure that only authorized users can access backup data.

Cost Advantages

- No need to purchase expensive storage hardware.
- Pay only for the storage used (Pay-as-you-go model).
- Cloud providers manage maintenance, upgrades, and backups.

Disaster Recovery Example

If a company's data center is damaged due to fire or flood, backup data stored in the cloud can be restored quickly, ensuring business continuity.

Conclusion

Cloud storage provides **secure, durable, cost-effective, and reliable backup** with fast disaster recovery.

3. Global Mobile Application Deployment

How CDN Improves Performance

- A **Content Delivery Network (CDN)** stores copies of application content on multiple edge servers worldwide.
- Users receive content from the nearest server, reducing latency.

Role of Edge Servers

- Edge servers process user requests closer to their location.
- They reduce network congestion and improve application speed.

Global Availability

Cloud providers maintain data centers in multiple regions worldwide. If one region fails, another region continues providing services.

Real-Time Example

In **Netflix**, videos are delivered from nearby CDN servers to reduce buffering.

In **online gaming**, game updates and assets are served from nearby edge servers to reduce lag.

Conclusion

CDNs improve **performance, reduce latency, increase availability, and enhance user experience globally.**

4. Financial Institution – Security and Compliance

Regulatory Compliance and Auditing

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- GDPR
- SOC 2

These certifications help organizations meet security and compliance requirements.

Role of Logging, Monitoring, and Encryption

- **Logging:** Records every user activity for auditing.
- **Monitoring:** Detects suspicious activities and security threats.
- **Encryption:** Protects sensitive financial data during storage and transmission.

Ensuring Data Sovereignty

- Store financial data only in approved geographic regions.
- Use region-specific cloud data centers to comply with government regulations.

Example

A bank stores customer transaction records in an Indian cloud region with AES-256 encryption, MFA, audit logs, and continuous monitoring to meet RBI and compliance requirements.

Conclusion

Cloud security features help financial institutions maintain **compliance, confidentiality, integrity, and secure financial transactions.**

5. Containerization and Kubernetes

Containers vs Virtual Machines

Containers

Share the host operating system

Lightweight

Start in seconds

Lower resource usage

High portability

Virtual Machines

Each VM has its own operating system

Heavyweight

Takes minutes to boot

Higher CPU and memory usage

Less portable

How Kubernetes Helps

- Automates container deployment.
- Automatically scales applications based on traffic.
- Performs self-healing by restarting failed containers.
- Provides load balancing among containers.

Portability Benefits

Containers can run consistently across:

- Private Cloud
- Public Cloud
- Hybrid Cloud
- Developer laptops

Microservices Example

An **e-commerce application** consists of separate services for:

- User Authentication
- Product Catalog
- Shopping Cart

- Payment Service
- Order Management

Each service runs in its own container and is managed by Kubernetes.

Conclusion

Containerization provides **fast deployment, efficient resource utilization, portability, and simplified application management**, while Kubernetes automates deployment, scaling, and fault recovery.

These answers directly address all parts of the five application-based questions in your uploaded assessment and are suitable for a **5-mark university exam**.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity